

CS & IT ENGINEERING



Digital Logic

Optimisation

Lecture No. 07



By- Aditya sir

Recap of Previous Lecture



Topic



K-maps (POS)

Topics to be Covered



Topic



k-maps

- Implicants
- Prime Implicants
- Essential Prime Implicants
- Selective PI → Reduced PI.

1) ~~Implicants~~

2) ~~Prime Implicants (PI)~~

3) Essential Prime Implicants (EPI)

4) Selective Prime Implicants (SPI)

5) Reduced Prime Implicants (RPI)

(mandatory)
PI present in
every ans

PI present in
the ans but
are not EPI

PI that are not
EPI and not SPI

eg 1:- $f(P, Q, R) = \sum m(2, 3, 4, 5, 7)$

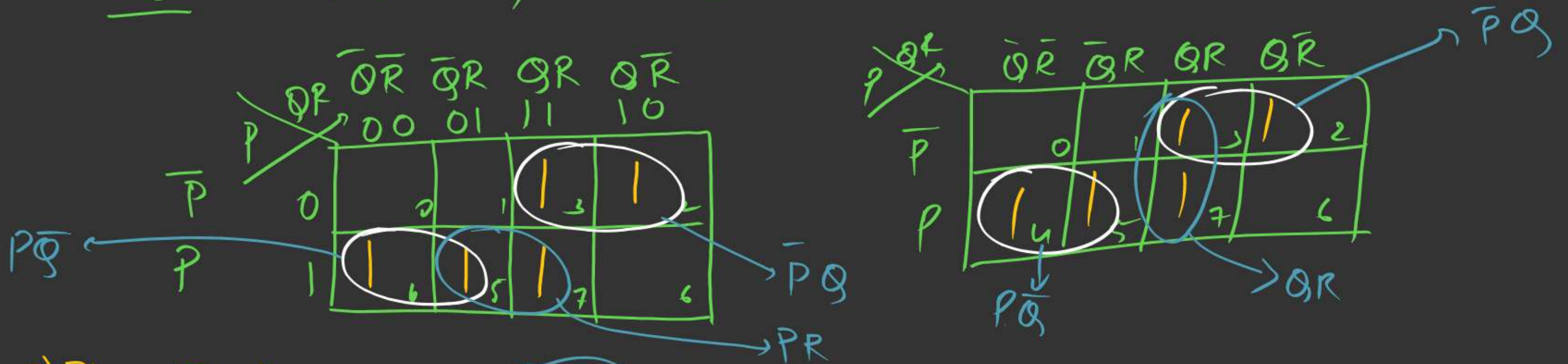
		$\overline{Q}\overline{R}$	$\overline{Q}R$	QR	$Q\overline{R}$
		00	01	11	10
$\overline{P}$	0	0	1	1	1
P	1	1	1	1	0

① Implicants: The no. of min-terms in the Boolean expression.

OR $\Rightarrow$ The total no. of 1s in the K-map.

Implicants = 5

eg 1:- $f(P, Q, R) = \sum m(2, 3, 4, 5, 7)$



Prime Implicants: The total no. of possibilities of forming the groups in K-map. $PI = 4 \{ P\bar{Q}, \bar{P}Q, PR, QR \}$

largest possible (no subgroups)

eg1: $f(P, Q, R) = \sum m(2, 3, 4, 5, 7)$

P \ QR	$\bar{Q}\bar{R}$ $\bar{Q}R$ QR $Q\bar{R}$			
	00	01	11	10
$\bar{P}$	0	0	1	1
P	1	1	1	0

Annotations for the first K-map:
 - Group $\bar{P}\bar{Q}$ (cells 2, 3)
 - Group $\bar{P}Q$ (cells 3, 7)
 - Group $P\bar{Q}$ (cells 4, 5)
 - Group PR (cells 3, 5, 7)

P \ QR	$\bar{Q}\bar{R}$ $\bar{Q}R$ QR $Q\bar{R}$			
	00	01	11	10
$\bar{P}$	0	0	1	1
P	1	1	1	0

Annotations for the second K-map:
 - Group $\bar{P}\bar{Q}$ (cells 2, 3)
 - Group $\bar{P}Q$ (cells 3, 7)
 - Group $P\bar{Q}$ (cells 4, 5)
 - Group QR (cells 3, 7)

Prime Implicants
 = $\{ \check{P\bar{Q}}, \check{PR}, \check{\bar{P}Q}, \check{QR} \}$
 (4)

ans

$$\underline{\underline{F = P\bar{Q} + \bar{P}Q + PR}}$$

$$F = \underline{\underline{P\bar{Q} + \bar{P}Q + QR}}$$

③ Essential

Prime Impl: $\{ \underline{\underline{P\bar{Q}}}, \underline{\underline{\bar{P}Q}} \} = \underline{\underline{2}}$

SPI = 1

RPI = 1

Q92: $f(P, Q, R) = \overline{P}\overline{Q}\overline{R} + \overline{P}\overline{Q}R + \overline{P}QR + P\overline{Q}\overline{R} + PQR$

$\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 0 & 1 & 3 & 6 & 7 \end{array}$

$= \sum m(0, 1, 3, 6, 7)$

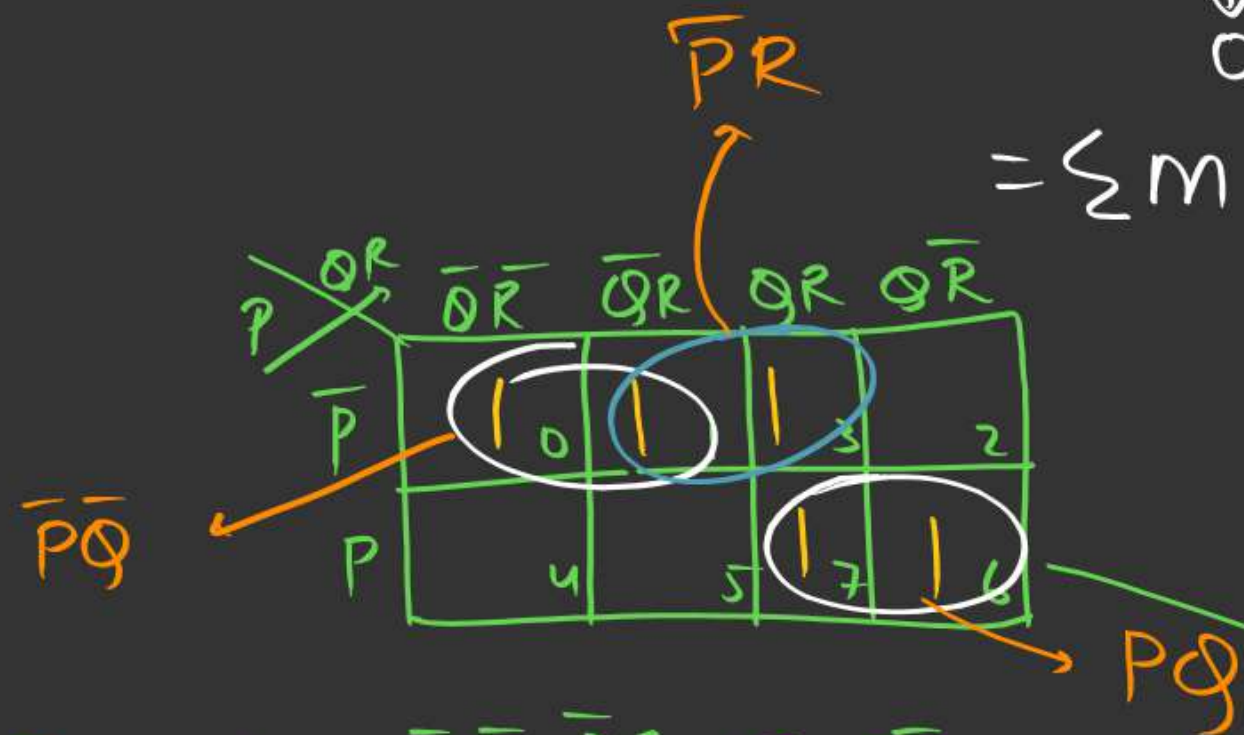
$P \backslash QR$	$\overline{Q}\overline{R}$	$\overline{Q}R$	QR	$Q\overline{R}$
$\overline{P}$	1 0	1 1	1 3	2
P	4	5	1 7	1 6

Implicants = 5

Q2: $f(P, Q, R) = \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}QR + P\bar{Q}\bar{R} + PQR$

$\begin{matrix} 0 & 0 & 0 \\ \downarrow & & \\ 0 & & \end{matrix}$
 $\begin{matrix} 0 & 0 & 1 \\ \downarrow & & \\ 1 & & \end{matrix}$
 $\begin{matrix} 0 & 1 & 1 \\ \downarrow & & \\ 3 & & \end{matrix}$
 $\begin{matrix} 1 & 1 & 0 \\ \downarrow & & \\ 6 & & \end{matrix}$
 $\begin{matrix} 1 & 1 & 1 \\ \downarrow & & \\ 7 & & \end{matrix}$

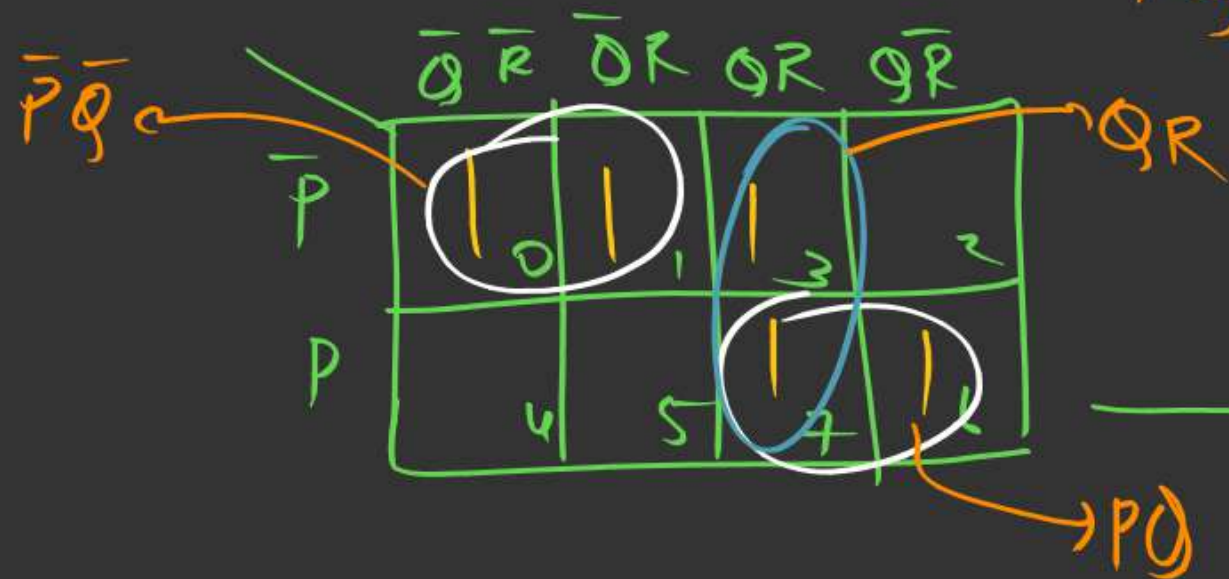
$= \sum m(0, 1, 3, 6, 7)$



Implicants = 5

Prime Implicants = $\{\bar{P}\bar{Q}, PQ, \bar{P}R, QR\}$

$f = \bar{P}\bar{Q} + PQ + \bar{P}R = 4$



$f = \bar{P}\bar{Q} + PQ + QR$

(EPI)

③ Essential PI
= 2 $\{\bar{P}\bar{Q}, PQ\}$

SPI = 1
RPI = 1

Q3 $f(P, Q, R) = \sum m(0, 1, 3, 4, 6, 7)$

Implicants = 1

Prime Implicants = $\{\bar{Q}\bar{R}, \bar{P}R, PQ, P\bar{R}, QR, \bar{P}\bar{Q}\} = 6$

$\bar{Q}\bar{R}$ $\bar{P}R$

	$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	1 ₀	1 ₁	1 ₃	
P	1 ₄		1 ₇	1 ₆

PQ

$f = \bar{Q}\bar{R} + \bar{P}R + PQ$

$\bar{P}\bar{Q}$ QR

	$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	1 ₀	1 ₁	1 ₃	
P			1 ₇	1 ₆

$P\bar{R}$

$f = \bar{P}\bar{Q} + QR + P\bar{R}$

$I = ?$
 $\bar{P}I = ?$

$(2, 7)$

$EPI = 0$

$SPI = 3$

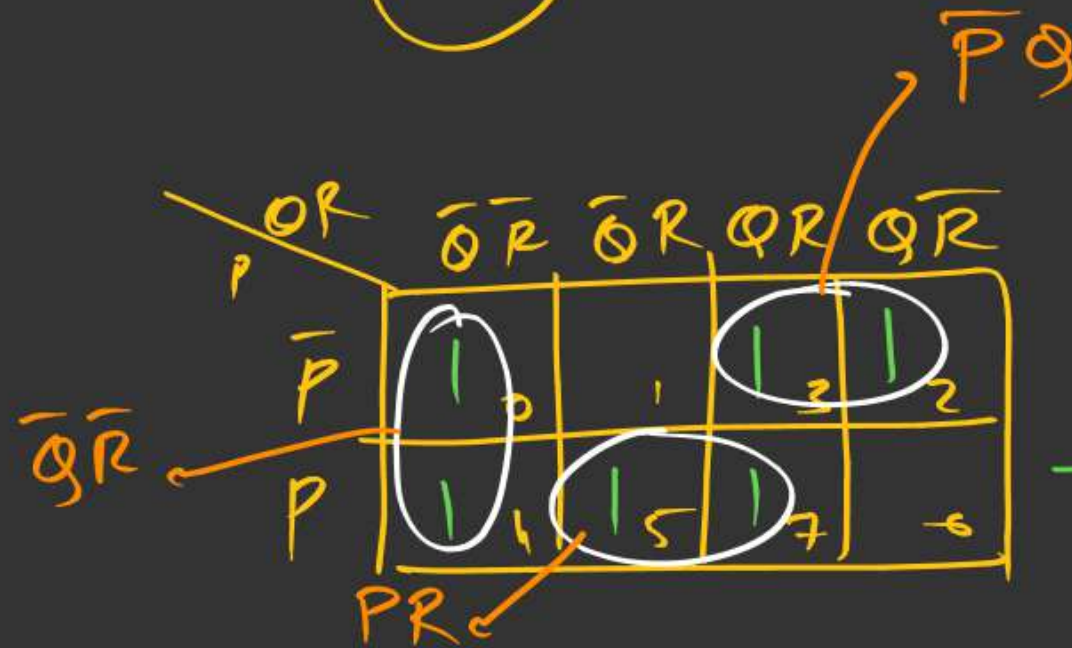
$RPI = 3$

(294)

$$f(P, Q, R) = \sum m(0, 2, 3, 4, 5, 7)$$

✓ 85%
(PI)

Implicants = 6

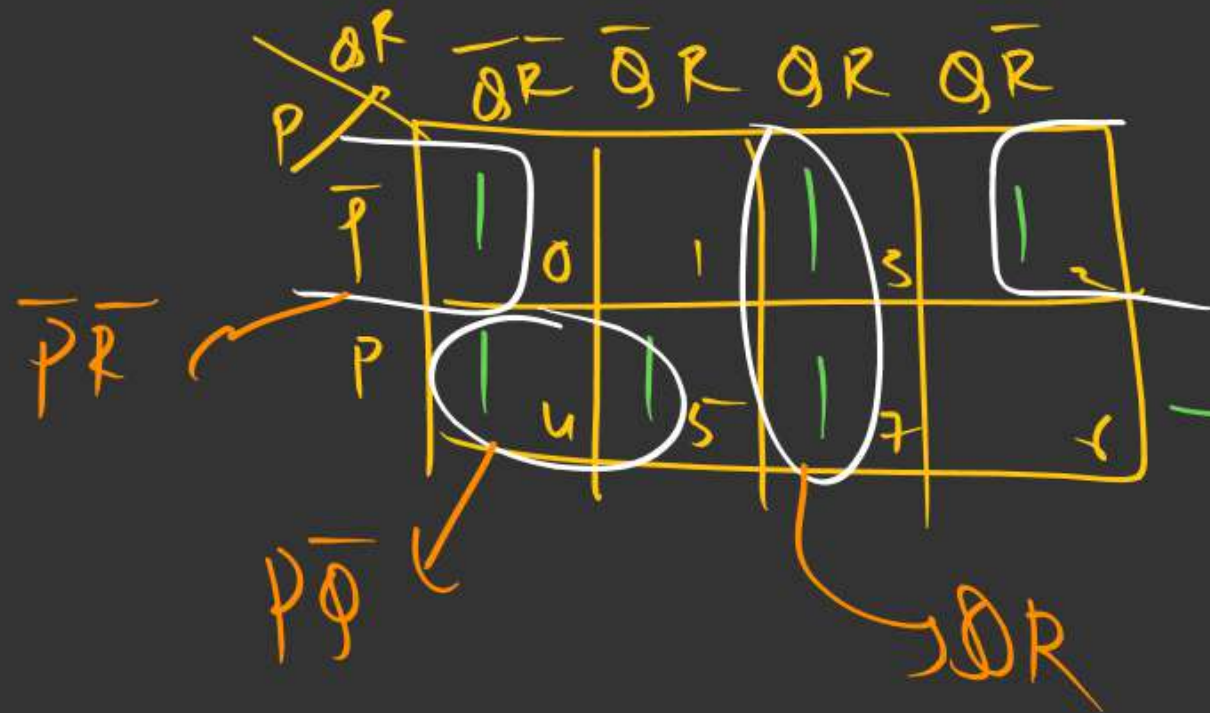


$$F = \overline{Q}R + \overline{P}Q + PR$$

$$EPI = 0$$

$$SPI = 3$$

$$RPI = 3$$



$$F = \overline{P}R + P\overline{Q} + QR$$

Prime Implicants

$$= \{ \overline{Q}R, PR, \overline{P}Q, \overline{P}R, P\overline{Q}, QR \} = 6$$

V. Imp eg 5)

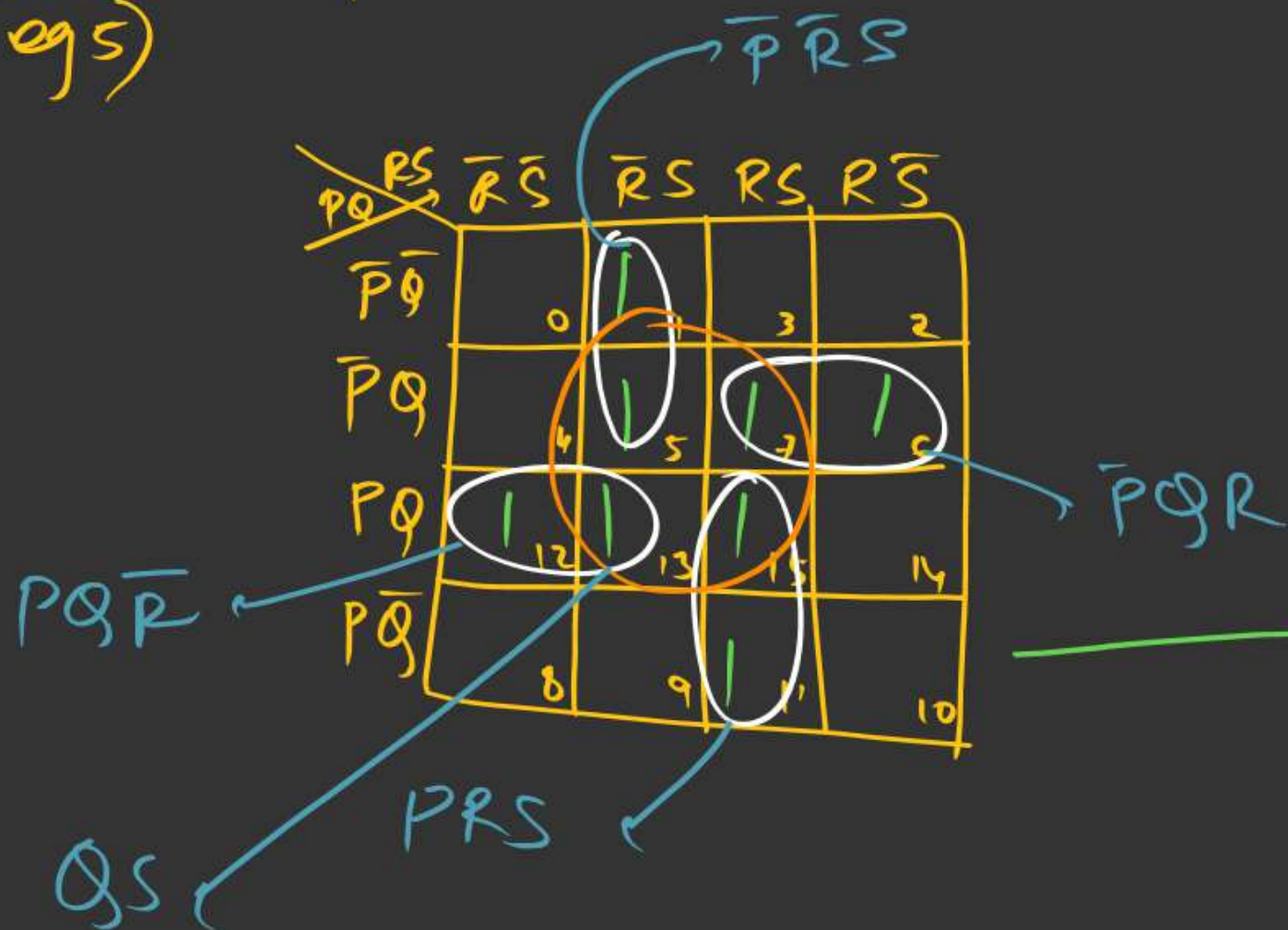
Challenge: $f(P, Q, R, S) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

75%

PI = ?

80%

↓
30%



SOP = $[PQ\bar{R} + \bar{P}\bar{R}S + \bar{P}QR + PRS]$
(7)

EPI = 4

SPI = 0

RPI = 1

Prime Implicants = $\{PQ\bar{R}, \bar{P}\bar{R}S, \bar{P}QR, PRS, QS\}$
not an EPI PRS, QS

$$(Q) \textcircled{6} f(P, Q, R) = \sum m(0, 1, 3, 5, 6, 7)$$

$\bar{P}\bar{Q}$ $\bar{P}Q$ $P\bar{Q}$ PQ

	$\bar{Q}\bar{R}$	$\bar{Q}R$	QR	$Q\bar{R}$
$\bar{P}$	1 0	1 1	1 3	2
P	4	1 5	1 7	1 6

R PQ

Prime Impl: $\{ \bar{P}\bar{Q}, R, PQ \}$

= 3

EPI = 3

$|Impl| = 6$

PI

EPI

Telegram

SPI = 0

RPI = 0

④

Q. 7) $f(P, Q, R, S) = \sum m(0, 1, 4, 5, 6, 10, 11, 13, 14, 15)$

$I = 10$

$\boxed{PI = 6}$

$EPI = 2$

$SPI = 2$

$RPI = 2$

$\bar{P}\bar{R}$

EPI

	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1 0	1 1		
$\bar{P}Q$	1 4	1 5		1 6
$P\bar{Q}$		1 12	1 13	1 14
PQ			1 11	1 10

$Q\bar{R}\bar{S}$

$\bar{P}Q\bar{S}$

$PI = \{ \bar{P}\bar{R}, \underline{PR}, Q\bar{R}\bar{S}, PQS, \bar{P}Q\bar{S}, Q\bar{R}\bar{S} \}$

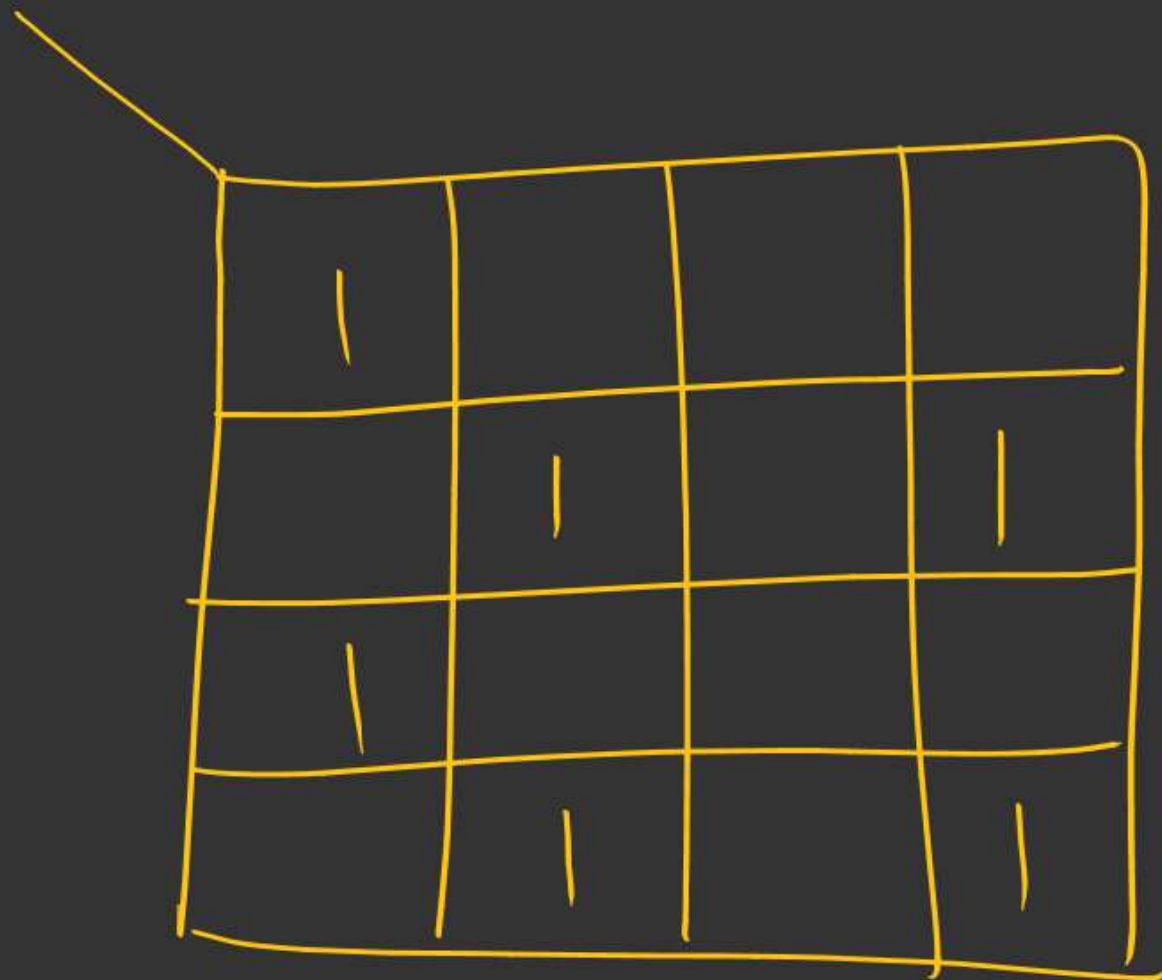
$PQ\bar{S}$

$Q\bar{R}\bar{S}$

PR

EPI

HW:-



1			
	1		1
1			
	1		1





2 mins Summary



— I
→ PI
→ EPI
→ SPI
→ RPI



THANK - YOU

